



Smart glasses that decode facial expressions can help children with autism develop better social skills (Source: Steve Fisch and Stanford Medicine News Center)

consistent practice and feedback, adapt to individual learning paces, mimic everyday social conversations, and offer predictive text and communication boards that learn an individual's patterns, enabling them to learn at their own pace without any pressure. AI-powered cameras can recognise objects and provide spoken labels to expand vocabulary. Speech recognition and generation systems convert verbal attempts into text and generate speech for individuals using alternative communication methods.

Proloquo2Go, TippyTalk, Avaz AAC, Jellow AAC, and QuickPic AAC are some examples of AAC tools. Others like Otsimo and InnerVoice help neurodivergent people to practice conversations and social interactions. A winner of Microsoft's *AI for Accessibility* grant, iTherapy's InnerVoice includes a lot of AI-powered features built on the Azure platform. Its *visual language* feature uses AI innovatively to teach language. When you see something, the camera takes a picture, and the AI system labels

the picture with text and describes it with speech, indirectly establishing the relationship between objects in the environment, speech, language, and text. It teaches users social communication skills using animated 3D *avatars* of themselves, or their favourite toys, and cartoon characters. Emojis and *avatars* come to life to teach the children how to emote.

Content creation. Parents and therapists can use generative AI to quickly create personalised social stories—text, audio, and videos—that help children understand and prepare for new or potentially stressful situations. Depending on the means and availability, some therapists have also started using immersive virtual reality (VR) and augmented reality (AR) content for personalised life skills coaching. However, the content must be carefully curated based on the learner's emotional forbearance.

Adaptive learning. Personalisation and adaptability are key to the success of special education. AI-powered tools help educators

go one step ahead of personalisation, by constantly monitoring the learning experience and adjusting educational content and pace in real time based on the neurodivergent learner's strengths, challenges, and progress. They increase accessibility to learning materials by dynamically simplifying them or presenting information in different formats that suit the user. This helps prevent frustration, boosts confidence, and keeps the child engaged. Otsimo, an AI-powered platform for special education and speech therapy, for example, offers engaging, interactive games that automatically adjust content, pace, and activities to a child's unique learning style and progress.

Social robots and AI companions. Robots such as NAO, Milo, Jibo, and Kaspar help teach social skills to those with ASD. They can recognise facial expressions and adapt their responses to the person's emotional state and learning progress. Friendly and patient beyond our wildest imagination, these social robots provide a very safe, consistent, and stress-free environment for neurodivergent individuals of all ages to practice social cues. Some AI-powered mental health chatbots like Wysa may help neurodivergent individuals deal with anxiety and burnout—but these are more for adult users, and should be used with caution, because the suggestions may not be appropriate for the individual's specific needs and challenges. Caregivers might not have control over the content and interactions.

Real-world experiences. The combination of AI, VR, and AR technologies is highly effective in helping children and adults acquire essential life skills. VR and AR platforms simulate customised,